

# QUBO Formulation: Binary Image Denoising

## Binary Image Denoising

**Problem Statement.** Given a noisy binary image represented on a grid of  $N$  pixels, the objective is to recover a clean binary image that minimizes a global energy function. This energy comprises two components: a data fidelity term that penalizes deviations between the reconstructed pixels and the observed noisy pixels, and a smoothness term that penalizes label discontinuities between spatially adjacent pixels. The task is to determine the binary configuration that achieves the minimum total energy.

**Decision Variables.** Let  $\mathcal{K} = \{0, 1, \dots, N-1\}$  denote the set of indices corresponding to the pixels in the image grid. For each  $k \in \mathcal{K}$ , we introduce a binary decision variable  $x_k \in \{0, 1\}$ , where  $x_k = 1$  indicates a foreground (active) pixel and  $x_k = 0$  indicates a background (inactive) pixel in the reconstructed clean image. The vector of all variables is denoted  $\mathbf{x} = (x_0, x_1, \dots, x_{N-1})^\top$ .

**QUBO Derivation.** We derive the Quadratic Unconstrained Binary Optimization (QUBO) formulation through the following algebraic steps.

**Step 1.** The original objective is expressed as a minimization over the binary hypercube:

$$\min_{\mathbf{x} \in \{0,1\}^N} \sum_{k \in \mathcal{K}} d_k(x_k) + \sum_{(k,l) \in \mathcal{E}} w_{kl} |x_k - x_l|, \quad (1)$$

where  $\mathcal{E}$  is the set of adjacent pixel pairs,  $w_{kl} \geq 0$  are smoothness weights, and  $d_k(x_k)$  is the instance-dependent data cost for pixel  $k$ .

**Step 2.** The problem formulation contains no explicit equality or inequality constraints. Consequently, the constraint set is empty, and no constraint equations are introduced.

**Step 3.** Since there are no constraints, penalty terms are omitted. We proceed to convert the objective terms into a quadratic polynomial using the idempotent property of binary variables,  $x_k^2 = x_k$ . For binary variables, the absolute difference simplifies to:

$$|x_k - x_l| = x_k + x_l - 2x_k x_l. \quad (2)$$

The data cost  $d_k(x_k)$  can be represented by its linear coefficient  $\delta_k$  such that  $d_k(x_k) = \delta_k x_k$  (constant offsets are absorbed into the global offset). Substituting these expansions into the

objective yields:

$$H(\mathbf{x}) = \sum_{k \in \mathcal{K}} \delta_k x_k + \sum_{(k,l) \in \mathcal{E}} w_{kl} (x_k + x_l - 2x_k x_l). \quad (3)$$

**Step 4.** We combine and collect terms by variable index. Each quadratic term  $x_k x_l$  appears exactly once in the edge sum. Each linear term  $x_k$  accumulates contributions from its own data cost and from every incident edge  $(k, l) \in \mathcal{E}$ :

$$H(\mathbf{x}) = \sum_{k \in \mathcal{K}} \left( \delta_k + \sum_{l \in \mathcal{N}(k)} w_{kl} \right) x_k - \sum_{(k,l) \in \mathcal{E}} 2w_{kl} x_k x_l, \quad (4)$$

where  $\mathcal{N}(k) = \{l \mid (k, l) \in \mathcal{E}\}$  denotes the set of neighbors of pixel  $k$ .

**Step 5.** Reading off the coefficients from the expanded Hamiltonian  $H(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c$ , we obtain the general closed-form expressions for the QUBO parameters:

$$Q_{k,k} = \delta_k + \sum_{l \in \mathcal{N}(k)} w_{kl}, \quad \forall k \in \mathcal{K}, \quad (5)$$

$$Q_{k,l} = -2w_{kl}, \quad \forall (k, l) \in \mathcal{E}, \quad (6)$$

$$Q_{k,l} = 0, \quad \forall (k, l) \notin \mathcal{E}, k \neq l, \quad (7)$$

$$c = 0. \quad (8)$$

**Penalty Weight Justification.** The problem definition specifies no explicit constraints; therefore, penalty terms and associated penalty weights are omitted from the formulation. The optimization is strictly unconstrained over  $\{0, 1\}^N$ . If regularization or hard constraints were introduced in a variant of this problem, the penalty weight  $\lambda$  would be chosen to satisfy  $\lambda > \max_{\mathbf{x}} |H_{\text{obj}}(\mathbf{x})|$ , ensuring that any violation of feasibility incurs an energy penalty strictly larger than the maximum possible reduction in the objective function.

**Correctness Argument.** Minimizing  $H(\mathbf{x})$  over  $\{0, 1\}^N$  exactly recovers the optimal clean image because the QUBO is an algebraically equivalent reformulation of the original energy function. The expansion using  $x_k^2 = x_k$  preserves the exact numerical value of the objective for every binary configuration, meaning the relative ordering of all  $2^N$  candidate solutions remains unchanged. Since the problem is unconstrained, every binary vector is feasible, and the global minimum of  $H(\mathbf{x})$  corresponds precisely to the configuration that minimizes the combined data fidelity and smoothness costs. The quadratic structure ensures the energy landscape is well-defined for both classical and quantum solvers without introducing artificial feasible regions or penalty-induced local minima.

**Illustrative Example.** We instantiate the general formulation for a small  $2 \times 2$  image grid, yielding  $N = 4$  pixels indexed by  $\mathcal{K} = \{0, 1, 2, 3\}$ . The adjacency set is  $\mathcal{E} = \{(0, 1), (0, 2), (1, 3), (2, 3)\}$ . Assigning instance parameters  $d_k = \delta_k = 1$  for all  $k$  and uniform

smoothness weights  $w_{kl} = 1$  for all  $(k, l) \in \mathcal{E}$ , we substitute these values into the general QUBO expressions.

The concrete objective terms become:

$$\text{Linear: } Q_{0,0} = 1 + 1 + 1 = 3, \quad Q_{1,1} = 1 + 1 + 1 = 3, \quad (9)$$

$$Q_{2,2} = 1 + 1 + 1 = 3, \quad Q_{3,3} = 1 + 1 + 1 = 3. \quad (10)$$

$$\text{Quadratic: } Q_{0,1} = -2(1) = -2, \quad Q_{0,2} = -2(1) = -2, \quad (11)$$

$$Q_{1,3} = -2(1) = -2, \quad Q_{2,3} = -2(1) = -2. \quad (12)$$

The resulting QUBO Hamiltonian is:

$$H(\mathbf{x}) = 3(x_0 + x_1 + x_2 + x_3) - 2(x_0x_1 + x_0x_2 + x_1x_3 + x_2x_3). \quad (13)$$

Enumerating all  $2^4 = 16$  binary configurations reveals that the global minimum occurs at  $\mathbf{x}^* = (0, 0, 0, 0)^\top$  with minimum energy  $H(\mathbf{x}^*) = 0$ . This configuration corresponds to a completely uniform background image, which is optimal under the chosen parameters since the smoothness penalty for any transition outweighs the data cost savings. For other parameter regimes (e.g., larger  $d_k$  relative to  $w_{kl}$ ), the optimal solution would flip specific bits to match the noisy observations while preserving local smoothness.